

Linear regression

International Marketing Analytics



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[Office hours: book online \(Google Calendar\)](#)

Theory

Tests

Practice

Python

In class

Exercise

Understand



- What OLS does and what a coefficient means
- t-test, p-value, confidence interval, R^2
- Logs and elasticities

Check



- Seven assumptions behind OLS
- Residual plots and four tests
- What to do when a test fails

Code

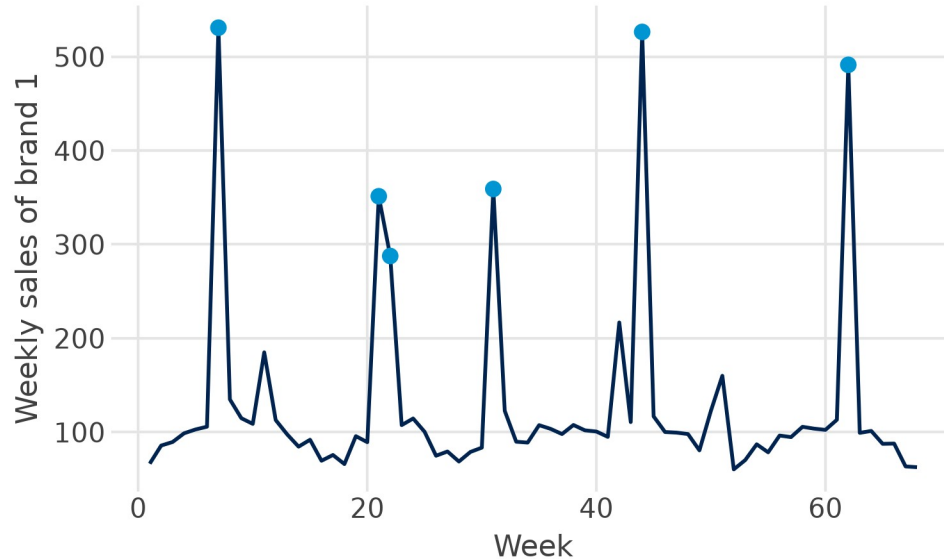


- A first regression with statsmodels
- Simple charts with plotnine
- A pricing recommendation from data



Readings: Bojinov, Parzen & Hamilton (2025), Linear Regression (HBS); Skiera, Reiner & Albers (2022), Regression Analysis.

The question: what does a price cut sell?



Weekly sales of brand 1; blue: weeks with a price cut (price below 1.20).

Data: 68 weeks of real scanner data, 4 brands

Normal week: price about 1.50, sales about 100

Price-cut week: price about 1.03, sales up to 530

Also in the data: competitor prices, features, displays, temperature, holidays



The manager asks: how much does 1 % lower price sell, and is the regular price right?

Theory



Part 1 of 6 · correlation, simple and multiple regression, inference, fit, logs, from elasticity to price

Correlation: direction and strength

Rule of thumb for $|r|$

$ r $	Interpretation
0 to 0.2	very weak
0.2 to 0.4	weak to moderate
0.4 to 0.6	medium to substantial
0.6 to 0.8	very strong
0.8 to 1.0	extremely strong

-0.94 sales and own price of brand 1:
an extremely strong negative
relationship

Linear only: a perfect U-shape has $r = 0$

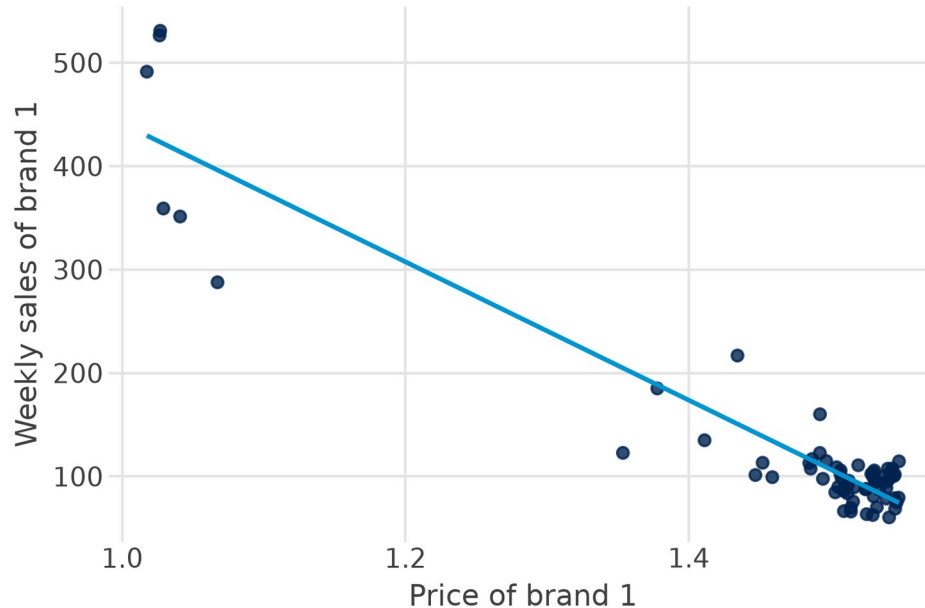
Not causation: ice cream and sunburn rise
together; the sun drives both



Correlation describes; regression quantifies: how many units does one euro of price cost?

Sources: Bojinov, Parzen & Hamilton 2025 (HBS note 9-622-100)

Simple linear regression



$$\text{sales} = b_0 + b_1 \cdot \text{price1} + e$$

$b_0 = 1,110$: intercept, sales at price 0
(not meaningful here)

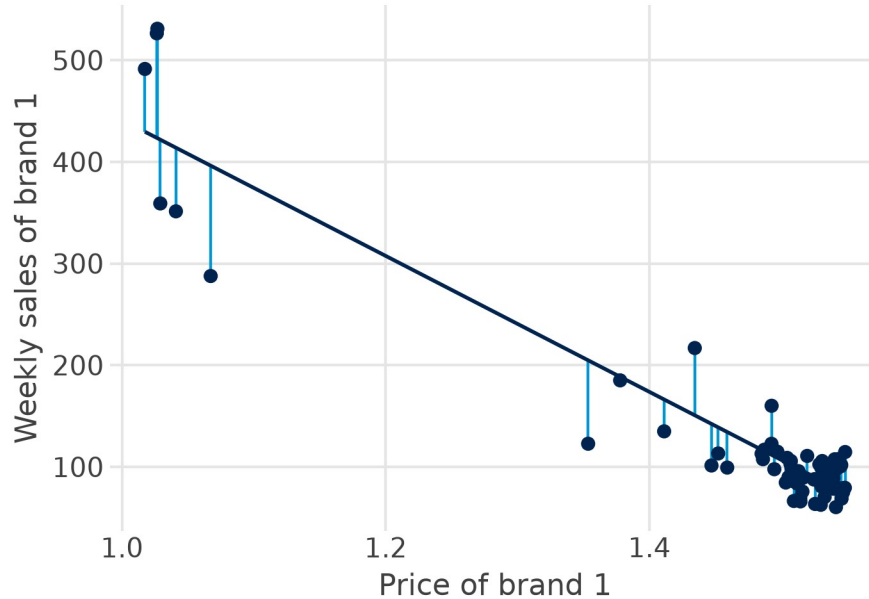
$b_1 = -669$: 10 cents more price, about 67
units fewer a week

e : residual, what the line does not explain



Look at the chart: the points curve, so a straight line misses the price-cut weeks.

Least squares: the line with the smallest squared errors



residual: $e_i = y_i - \hat{y}_i$
OLS: minimise $\sum e_i^2$

Squared: misses above and below do not cancel

Large misses count much more than small ones

One line: the unique best fit for these data

Sources: Skiera, Reiner & Albers 2022; Bojinov, Parzen & Hamilton 2025

Multiple regression: holding other things constant

$$\ln \text{ sales} = b_0 + b_1 \cdot \ln \text{ price}_1 + b_2 \cdot \ln \text{ price}_2 + b_3 \cdot \ln \text{ price}_3 + b_4 \cdot \ln \text{ price}_4 \\ + b_5 \cdot \text{feature}_1 + b_6 \cdot \text{temp} + b_7 \cdot \text{december} + e$$

Own mix

Price of brand 1
Feature ads, displays

Competitors

Prices of brands 2 to 4
Their features and displays

Context

Temperature
December and Easter weeks



Each b: the effect of one variable with all others held constant (*ceteris paribus*).

Left out a relevant variable? Its effect ends up in the others (omitted variable bias).

Is the effect real? t-test, p-value, interval, F-test

Σ t-test

$$t = b / se$$

Own price: $-3.64 / 0.30 = -12.1$

p-value

Below 0.05: significant

Own price: $p < 0.001$; brand 4: $p = 0.86$

95 % confidence interval

About $b \pm 2 se$

Own price: -4.24 to -3.04

F-test

Are all slopes together zero?

$F = 59.5$, $p < 0.001$



Significant is not important: read the size of b and its interval, not just the stars.

Sources: Bojinov, Parzen & Hamilton 2025; Skiera, Reiner & Albers 2022

How well does the model fit?

0.874

R^2 : the model explains 87 % of the weekly variation in log sales

0.859

adjusted R^2 : R^2 with a penalty for each extra variable

0.169

RMSE: the typical weekly miss, about 17 % in a log model

Careful



- R^2 never falls when you add a variable
- Compare R^2 only for the same outcome (sales vs log sales)
- High R^2 does not mean causal



Main model for brand 1: log-log, own and competitor prices, feature, temperature, December; 68 weeks.

Sources: Bojinov, Parzen & Hamilton 2025; Skiera, Reiner & Albers 2022

Functional forms and dummy variables

Model	Equation	b means	Brand 1
Linear	$\text{sales} = a + b \cdot \text{price}$	+1 unit price: b units	$b = -669$ units
Log-log	$\ln \text{sales} = a + b \cdot \ln \text{price}$	+1 % price: b %	$b = -3.64$ (elasticity)
Log-linear	$\ln \text{sales} = a + b \cdot \text{temp}$	+1 °C: about $100 \cdot b$ %	$b = -0.0066$: -0.66 %
Dummy in log model	$\ln \text{sales} = a + b \cdot \text{december}$	$e^b - 1$, vs other weeks	$b = -0.385$: -32 %



Why logs? sales respond in per cent, not in units, and the elasticity can be compared across brands and countries.

Sources: Skiera, Reiner & Albers 2022; Wooldridge 2025, ch. 6 to 7

From elasticity to a pricing decision

optimal price $p^* = \text{marginal cost} \times \varepsilon / (1 + \varepsilon)$ (valid for $\varepsilon < -1$)

\$52

Skiera et al.: $\varepsilon = -2.34$, unit cost \$30 $\rightarrow p^* = 30 \times 2.34 / 1.34$

$\times 1.38$

Brand 1: $\varepsilon = -3.64 \rightarrow$ price 38 % above marginal cost

Short term: promotion weeks inflate the elasticity; regular-price elasticities are smaller

Competitors react, and the retailer takes its margin

Sources: Skiera, Reiner & Albers 2022; Hanssens, Parsons & Schultz 2001

Assumptions and tests



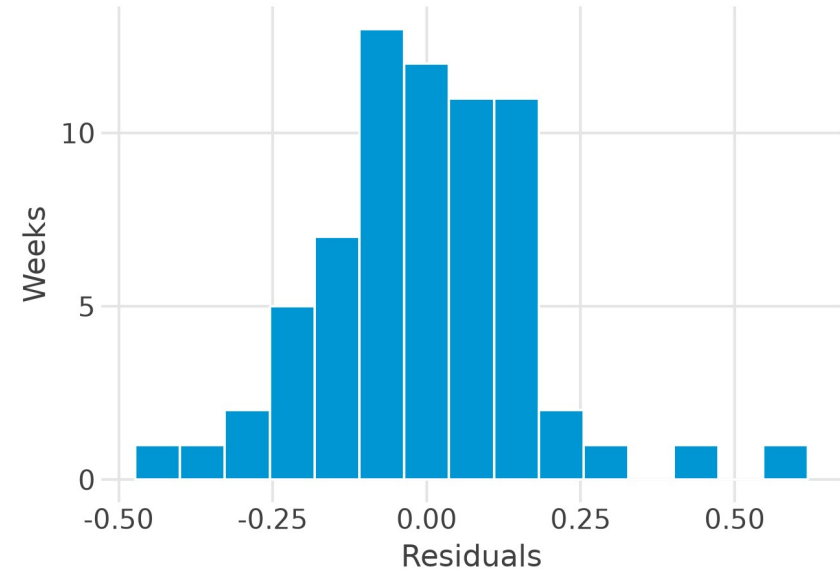
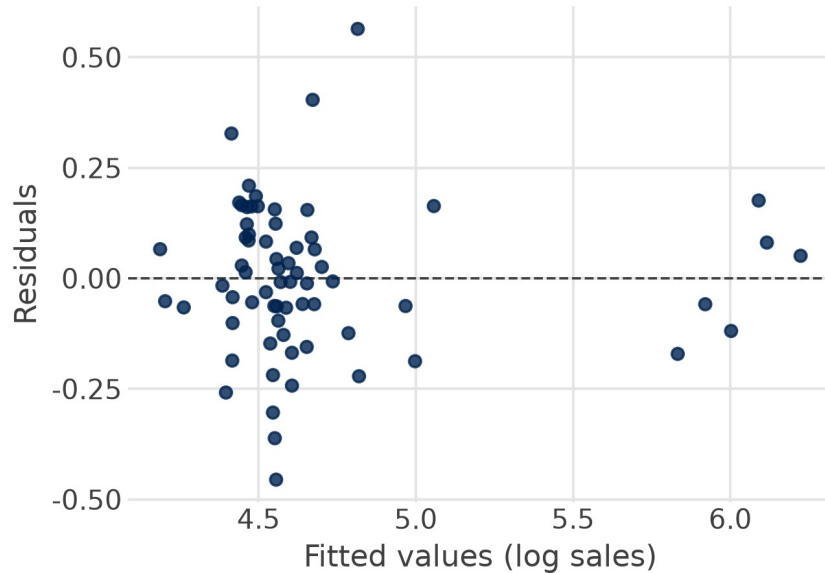
Part 2 of 6 · what OLS assumes, how to look,
how to test, what to do

Seven assumptions behind OLS

Assumption	If violated	Look or test (statsmodels)
1 Linear form	biased coefficients	residuals vs fitted; try logs
2 Exogeneity, no omitted variables	biased, not causal	theory, controls, experiments
3 Constant error variance	wrong standard errors	funnel in plot; Breusch-Pagan
4 No autocorrelation	standard errors too small	residuals over time; Durbin-Watson ≈ 2
5 Normal residuals	tests unreliable in small samples	histogram; Jarque-Bera
6 No multicollinearity	unstable b, large se	correlations; VIF below 10 (better 5)
7 Enough data, no dominant week	fragile results	$n \geq 3$ to $5 \times$ coefficients; Cook's distance

Sources: Skiera, Reiner & Albers 2022; Hair et al. 2019; Wooldridge 2025

Look first: two residual plots



Look for: a curve (wrong form) or a funnel (unequal variance)

Look for: a rough bell shape, no extreme weeks



Brand 1: no curve, no funnel, a roughly bell-shaped histogram.

Then test: four numbers for the brand 1 model

1.41

Durbin-Watson: below 2, some autocorrelation between weeks

0.14

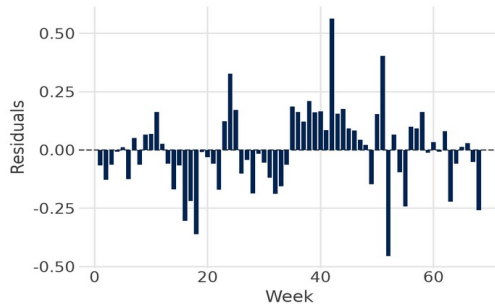
Breusch-Pagan p: no evidence of unequal variance

0.07

Jarque-Bera p: residuals roughly normal

2.4

largest VIF: no multicollinearity



Verdict: usable. Report Newey-West (HAC) standard errors because of the autocorrelation: 0.32 instead of 0.30 for own price.

When a test fails: what to do

Problem	Remedy	In statsmodels
Curve in residuals	logs, squares, interactions	<code>np.log(x), l(x**2)</code>
Unequal variance	robust standard errors	<code>fit(cov_type="HC3")</code>
Autocorrelation	HAC standard errors; lags; ARIMA errors (Session 4)	<code>fit(cov_type="HAC", cov_kwds={"maxlags": 1})</code>
Multicollinearity	more variation, combine variables, experiments	<code>variance_inflation_factor</code>
Dominant weeks	check data, report with and without	<code>get_influence().cooks_distance</code>
Endogeneity	experiments, instruments, fixed effects	Sessions 2 to 5

Sources: Skiera, Reiner & Albers 2022; statsmodels documentation

Three warnings from real data

Displays



- Only 4 display weeks
- VIF 11.4 and 10.9
- Effects cannot be separated

One Easter week



- A dummy for one week fits it exactly
- Leverage 1.00
- No real estimate of Easter

Mailings



- Skiera et al.: mailings and salespersons $r = 0.989$
- VIF 53
- Salesperson effect flips sign



The real fix: vary the marketing instruments independently, ideally in a field experiment.

Sources: Skiera, Reiner & Albers 2022; own calculation on the chocolate data

Endogeneity: when marketing follows demand

The assumption: explanatory variables are unrelated to the error term

Marketing breaks it: budgets go where managers expect success, so spend and unexplained sales rise together

Brand 1: price cuts come with features, so the price effect partly measures the feature

Evidence: in 15 Facebook experiments, regression often overstated the true lift, sometimes 3 times



Remedies: experiments (A/B, geo tests), instrumental variables such as cost shifters, panel data with fixed effects.

Sources: Skiera, Reiner & Albers 2022; Gordon, Zettelmeyer, Bhargava & Chapsky 2019

Regression in the real world



Part 3 of 6 · benchmarks, companies, what transfers to MMM

Benchmarks from decades of regressions

-2.62 mean price elasticity, 1,851 estimates (Bijmolt et al. 2005)

0.12 mean short-term advertising elasticity (Sethuraman et al. 2011)

-0.42 mean price elasticity in 14 Asia-Pacific countries, lower with high power distance (Datta et al. 2022)

-3.64 brand 1, chocolate: promotion-driven, many substitutes



Use benchmarks to judge whether your own estimate is plausible.

Where companies use regression

Meta Robyn

MMM in production:
ridge regression on media

Booking.com

With Microsoft: CUPED,
regression makes A/B tests
faster

Google

Geo experiments: test vs
control regions by regression

Walmart (M5)

Sales forecasts need price,
promotion and event
regressors

statworx

Frankfurt consultancy: price
elasticities, log-log

Bayer

Budget across countries
from estimated elasticities



Session 2: the marketing mix model is this regression with adstock and saturation.

Sources: Robyn docs; Deng et al. 2013; Vaver & Koehler 2011; Makridakis et al. 2022; Fischer et al. 2011

The chocolate data in Python



Part 4 of 6 · load and plot, the first regression,
the log-log model, the checks

The data: 68 weeks of chocolate scanner data

Column	Meaning
week	1 to 68
sales	weekly unit sales of brand 1
price1 to price4	prices of brands 1 to 4
feature1 to feature4	share of stores with a feature ad, 0 to 1
display1 to display4	share of stores with an in-store display
fand1, fand3, fand4	share with feature and display together
temp	temperature, °C
december, easter	1 in December and Easter weeks

File: data/legacy/chocolate_dataset.csv

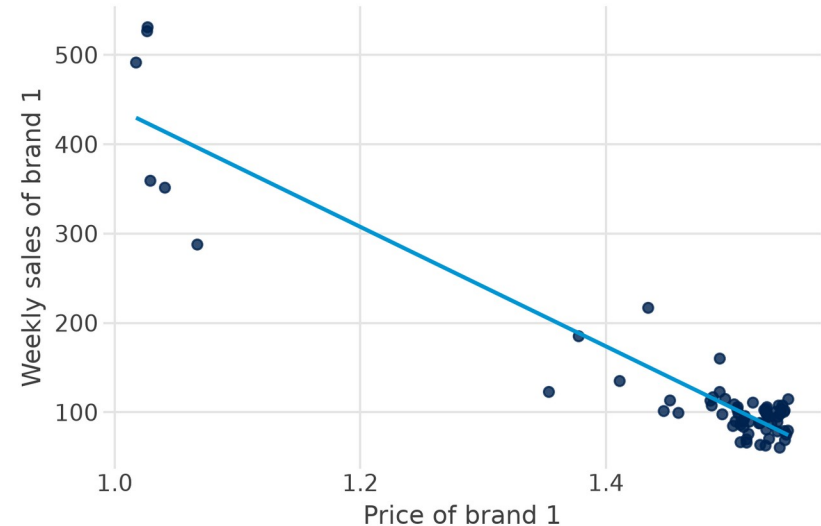
Step 1: load and look

Python: polars and plotnine

```
import polars as pl
from plotnine import (ggplot, aes, geom_point,
                      geom_smooth, labs)

choc = pl.read_csv(
    "data/legacy/chocolate_dataset.csv")
choc.select("sales", "price1",
            "price2").describe()

(ggplot(choc, aes("price1", "sales"))
 + geom_point()
 + geom_smooth(method="lm")
 + labs(x="Price of brand 1",
        y="Weekly sales"))
```



Look before you model: ranges, outliers, the shape of the relationship.

Step 2: the first regression

Python: statsmodels

```
import statsmodels.formula.api as smf

df = choc.to_pandas()
simple = smf.ols("sales ~ price1",
               data=df).fit()
print(simple.summary())

simple.params      # b0, b1
simple.rsquared    # 0.884
```

	coef	std err	P> t
Intercept	1,110.0	44.0	0.000
price1	-668.8	29.8	0.000

$R^2 = 0.884$, $n = 68$

Read it: 10 cents more price, about 67 units fewer per week; the price explains 88 % of the weekly variation.



But: the residuals curve. Next: the log-log model with the marketing mix.

Step 3: the log-log model with the marketing mix

Python: statsmodels

```
import numpy as np

model = smf.ols(
    "np.log(sales) ~ np.log(price1)"
    " + np.log(price2) + np.log(price3)"
    " + np.log(price4) + feature1"
    " + temp + december",
    data=df).fit()
print(model.summary())
```

	coef	p
ln price1	−3.64	0.000
ln price2	−0.80	0.076
ln price3	0.36	0.109
ln price4	−0.03	0.855
feature1	0.18	0.307
temp	−0.007	0.045
december	−0.38	0.009

$R^2 = 0.874$, adjusted 0.859, n = 68



Read it: 1 % higher price, 3.6 % fewer sales; December weeks 32 % lower; each degree warmer 0.7 % lower.

Step 4: check the assumptions

Python: tests and robust errors

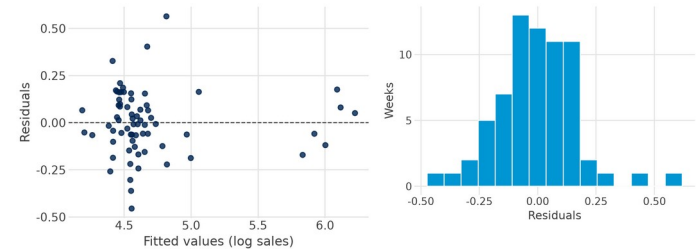
```
from statsmodels.stats.stattools import (
    durbin_watson, jarque_bera)
from statsmodels.stats.diagnostic import het_breuschpagan
from statsmodels.stats.outliers_influence import (
    variance_inflation_factor)

durbin_watson(model.resid)                # 1.41
het_breuschpagan(model.resid, model.model.exog)[1] # 0.14
jarque_bera(model.resid)[1]                # 0.07
X = model.model.exog
[variance_inflation_factor(X, i) for i in range(1, 8)]

hac = model.model.fit(cov_type="HAC",
                      cov_kwds={"maxlags": 1})
```

Residual plot

```
from plotnine import geom_hline
df["fit"] = model.fittedvalues
df["res"] = model.resid
(ggplot(df, aes("fit", "res"))
 + geom_point()
 + geom_hline(yintercept=0))
```



Decide: report the HAC result; the own-price elasticity stays -3.64 , its standard error rises to 0.32 .

In-class quiz: five questions

1 What does OLS minimise?

A the sum of residuals B the sum of absolute residuals C the sum of squared residuals

2 In $\ln \text{sales} = a + b \cdot \ln \text{price}$, $b = -3.64$ means ...

A 1 EUR more, 3.64 units fewer B 1 % higher price, 3.6 % fewer sales C 3.64 fewer per week

3 A VIF of 11 signals ...

A autocorrelation B multicollinearity C heteroscedasticity

4 Weekly data, Durbin-Watson = 1.41. This suggests ...

A positive autocorrelation B no problem C unequal variance

5 R^2 rises from 0.85 to 0.86 after adding a variable. This shows ...

A the variable matters B nothing: R^2 never falls C the model is causal



3 minutes alone, then compare with your neighbour. Same format as the online quiz and the exam.

In-class assignment



Part 5 of 6 · groups of 3, 25 minutes, on the chocolate data

In-class assignment: pricing brand 1

**Groups of 3:
20 minutes**

**Plenary:
5 minutes**

- 1 Competition:** plot sales against price2, price3 and price4. Which brand competes most?
- 2 Displays:** add display1 and fand1 to the log-log model. What happens to the VIFs and coefficients, and why?
- 3 Price:** marginal cost is 0.80. What price does $\epsilon = -3.64$ suggest? Why is the real price about 1.50?
- 4 The manager:** write one sentence for the brand manager, with a number and one caveat.

File: sessions/01-foundations/regression_chocolate.qmd (the code of Part 4 and these questions)

Coding exercise 1



Part 6 of 6 · individual, 5 % of the grade

Coding exercise 1: what to do

- 1 Explore:** a `great_tables` summary and two plotnine charts: sales over time, sales against price1
- 2 Simple model:** `sales ~ price1`; interpret the slope for a 10-cent price change
- 3 Log-log model:** own and competitor prices, `feature1`, `display1`, `temp`, `december`; interpret 3 coefficients
- 4 Assumptions:** residual plot, histogram, Durbin-Watson, Breusch-Pagan, Jarque-Bera, VIF; one sentence each
- 5 Robust errors:** refit with HAC standard errors; what changes and what does not?
- 6 Recommend:** the optimal price for marginal cost 0.90, with two caveats, in at most 100 words

Brief: `exercises/exercise-1.qmd` in the course repository and on Canvas

Coding exercise 1: submission and grading

Submit on Canvas



- exercise1_lastname.qmd and the rendered .html
- Renders from top to bottom without errors
- Every number in the text from code
- Prompt log at the end: what you asked, what you changed
- Deadline: see Canvas

Criterion	Points
Code runs, charts and tables clean	5
Interpretation of the coefficients	6
Assumption checks and conclusion	5
Recommendation with caveats	2
Prompt log	2
Total (5 % of the grade)	20



Individual work: discuss ideas, but write and check your own code. AI assistants are allowed; you own every number.

Key takeaways

- 1 OLS** draws the line with the smallest squared errors; each b holds the other variables constant
- 2 Logs** turn coefficients into elasticities: brand 1, -3.64
- 3 Look, then test:** residual plots, Durbin-Watson, Breusch-Pagan, Jarque-Bera, VIF
- 4 Fix what fails:** robust or HAC errors, better form, more variation; experiments for causality
- 5 Decide:** $p^* = \text{cost} \times \varepsilon / (1 + \varepsilon)$, with the caveats



Next session: the marketing mix model, a regression with adstock and saturation.

Academic



Bojinov, I. I., Parzen, M. & Hamilton, P. J. (2025). Linear Regression. HBS note 9-622-100 (Canvas).

Skiera, B., Reiner, J. & Albers, S. (2022). Regression analysis. In Homburg et al. (eds), Handbook of Market Research, 299–327. Springer.

Wooldridge, J. M. (2025). Introductory Econometrics, 8th ed. Cengage.

Hanssens, D. M., Parsons, L. J. & Schultz, R. L. (2001). Market Response Models. Kluwer.

Bijmolt, T., van Heerde, H. & Pieters, R. (2005). Journal of Marketing Research 42(2).

Gordon, B. R. et al. (2019). Marketing Science 38(2).

Learning resources



James, G. et al. (2023). ISLP, ch. 3 and the Python lab (free).

Schwarz, Chapman & Feit (2020). Python for Marketing Research and Analytics, ch. 7. Springer.

statsmodels. Documentation and example notebooks.

Turrell, A. Coding for Economists: regression and diagnostics (free).

StatQuest. Linear regression, clearly explained (video).

DataCamp. Introduction to Regression with statsmodels in Python.



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